INCH-POUND

MIL-L-40074E
29 March 1991
SUPERSEDING
MIL-L-40074D
12 July 1977

MILITARY SPECIFICATION

LEATHER, CALFSKIN AND KIP, FOR HEADBANDS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of full grain, vegetable tanned, perspiration resistant calfskin or kip leather for headbands.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8330

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SPECIFICATIONS

MILITARY

MIL-P-43432 - Packaging of Leather

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 22709.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Standard sample. The leather, when embossed, shall match the standard sample for fine hair cell pattern and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

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3.3. Material. The leather shall be whole skins of full grain, vegetable tanned calfskin or whole skins or sides of full grain, vegetable tanned kip. The flesh side of the leather shall be free of coarse loose flesh and shall be buffed smooth.

3.4 Trim (form) and area. The calfskin leather shall be furnished in the form of whole skins. The kip leather shall be furnished in the form of whole skins or sides. The area of the whole calfskin shall be not greater than 17-3/4 square feet. The area of the whole kipskin shall be not greater than 28 square feet. The area of the kip side shall be not greater than 14 square feet. The kip sides shall be trimmed to conform to the definition of a "side" as specified in FED-STD-311. The area expressed in square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.5 Thickness. The leather shall be 2 ounces $\pm$ 1/2 ounce in thickness when examined as specified in 4.4.3, and may be skived or shaved on the flesh side to insure greater uniformity of thickness.

NOTE: 1 ounce = 1/64 inch.

3.6 Color. The color of the leather shall be natural as produced by the vegetable tanning process.

3.6.1 Colorfastness. The leather shall be tested for resistance to dry and wet crocking in accordance with 4.4.4. Staining of the dry and wet crock cloths shall be not lower numerically than AATCC value 4.

3.7 Finish. The leather shall have a light application of colorless finish on the grain side. The finish shall contain no lead salts (see 6.5). A fine hair cell embossed pattern may be applied to the grain surface of the side. Printed leather shall not be permitted. If embossed, the leather shall match the standard sample for fine hair cell pattern.

3.8 Stitch tearing strength. At least 80 percent of the specimens tested shall have stitch tearing strength values of not less than 18 pounds when tested as specified in 4.4.4.

3.9 Area stability of leather to perspiration. At least 80 percent of the specimens tested shall have an area loss no greater than 15 percent (see 6.6) when tested as specified in 4.4.4.

3.10 Chemical requirements. The leather shall conform to the chemical requirements in table I when tested as specified in 4.4.4.

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TABLE I. Chemical requirements

Characteristic	Minimum	Maximum
Chloroform soluble materials, percent 1/	3.0	-
Lead (qualitative)	2/	-
pH value	3.3	5.0
Total ash, percent 1/	-	2.0

1/ Calculated on moisture-free basis.

2/ The presence of lead is prohibited.

3.11 Workmanship. The finished leather shall conform to the quality of product established by this specification, and the occurence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with the referenced documents.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of one calfskin, one kipskin, or one side as applicable. The sample unit shall be one calfskin, one kipskin, or one side as applicable. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for these examinations shall be performed in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determination on lots consisting of 51 or more calfskins, kipskins, or kip sides as applicable.

Examine	Defect
Material	Not calfskin or kip Not full grain Flesh side not buffed smooth or contains areas of loose flesh in excess of 4 square inches in total area
Trim	Not whole skin or side as applicable
Color	Color not natural as produced by tanning process
Finish	Grain side does not have colorless finish Embossing other than a fine hair cell pattern Printed design on leather Does not match the standard sample for fine hair cell pattern

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Examine	Defect
Quality of leather	Boardy or boney leather Discoloration, stain, cut, scratch, hole, brand, hipmark, rough shoulder, heavy vein, deep fat wrinkle, or other defects which individually or collectively impair the cutting value of more than 12 percent of the total area of the leather 1/
Marking	Omitted, illegible, or not in area specified

1/ Light or well healed scratches, well healed grub damage, light fat wrinkles, or slight stains shall not be classified as defects.

4.4.3 End item dimensional examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of sides. The sample unit shall be one side. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for this examination shall be performed in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more sample units.

Examine	Defect
Area expressed in terms of square feet	Not as marked on unit
Calfskin	Area greater than 17-3/4 square feet
Kip (whole skin)	Area greater than 28 square feet
Kip (side)	Area greater than 14 square feet
Thickness 1/	Not as specified

1/ Thickness shall be measured in accordance with Method 1011 of FED-STD-311 except that no individual thickness measurement shall be less than the minimum or greater than the maximum specified. Any sample unit having one or more thickness measurements outside the specified thickness range shall be classified as a defective unit.

4.4.4 End item testing. The methods of testing specified in FED-STD-311, wherever applicable and as listed in table II, shall be followed. Sampling procedures and location from which the sample unit is to be obtained shall be in

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accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exists:

a. More than three test specimens fail the stitch tearing strength requirements.
b. When tested for area stability, more than three specimens have an area loss greater than 15 percent.
c. More than one test failure occurs for the remaining requirements applicable to the sample unit.
d. Any composite test result fails to meet the specified requirement.

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite
Material identification	3.3	1/	-	-
Colorfastness	3.6.1	3031 2/	X	-
Stitch tearing strength	3.8	2151	X	-
Area stability of leather to perspiration	3.9	3211	X	-
Chloroform soluble materials	3.10	6341	-	X
Lead (qualitative)	3.10	6551	-	X
pH value	3.10	6621	-	X
Total ash	3.10	6421	-	X

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Color matching shall be done in indirect light.

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4.4.5 Packaging inspection. The sampling and inspection of the preservation, packaging, palletization, and marking shall be in accordance with the requirementss of MIL-P-43432.

5. PACKAGING

5.1 Preservation, packing, palletization and marking. The preservation, packing, palletization and marking for each level of protection shall be in accordance with the applicable requirements of MIL-P-43432 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The leather is intended for use in headbands.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1).
e. When palletization is required (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Lead salts. Lead chromate, lead pigments or other lead salts should not be used in the finish.

6.6 Glutaraldehyde treatment. Leather which has been treated with glutaraldehyde has been known to comply with the perspiration requirements of this specification.

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6.7 Subject term (key word) listing.

Full grain
Headwear
Hides
Vegetable tan

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 11

Preparing Activity:

Army - GL

(Project 8330-0189)

Review activities:

Army - MD
Air Force - 99, 82
DLA - CT

User activity:

Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-L-40074 E	2. DOCUMENT DATE (YYMMDD) 1991 March 29

3. DOCUMENT TITLE
Leather, Calfskin and Kip, For Headbands

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4532	(2) AUTOVON 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP, APR 90 *Previous editions are obsolete.* 198/29

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